Problem Set 3

AI 501: Mathematics for Artificial Intelligence | Fall 2026

Problem 1

Let

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & -1 \end{pmatrix}, \qquad B = \begin{pmatrix} -1 & 2 & 0 \\ 3 & 1 & 2 \end{pmatrix}.$$

Calculate:

(a) $A + B$;

(b) $3A - 2B$;

(c) A^T and B^T;

(d) $\|A\|_F$, the Frobenius norm of A.

Solution:

(a)

$$A + B = \begin{pmatrix} 0 & 0 & 3 \\ 3 & 5 & 1 \end{pmatrix}.$$

(b)

$$3A = \begin{pmatrix} 3 & -6 & 9 \\ 0 & 12 & -3 \end{pmatrix}, \qquad 2B = \begin{pmatrix} -2 & 4 & 0 \\ 6 & 2 & 4 \end{pmatrix}, \qquad 3A - 2B = \begin{pmatrix} 5 & -10 & 9 \\ -6 & 10 & -7 \end{pmatrix}.$$

(c)

$$A^T = \begin{pmatrix} 1 & 0 \\ -2 & 4 \\ 3 & -1 \end{pmatrix}, \qquad B^T = \begin{pmatrix} -1 & 3 \\ 2 & 1 \\ 0 & 2 \end{pmatrix}.$$

(d)

$$\|A\|_F = \sqrt{1^2 + (-2)^2 + 3^2 + 0^2 + 4^2 + (-1)^2} = \sqrt{31}.$$

Problem 2

Let

$$A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 3 & 4 \\ -2 & 0 & 5 \end{pmatrix}.$$

(a) Compute $\operatorname{tr}(A)$.

(b) Decompose $A = S + K$, where S is symmetric and K is skew-symmetric.

(c) Verify that $S^T = S$ and $K^T = -K$.

(d) Is A itself symmetric, skew-symmetric, or neither? Justify your answer using one of the parts above.

Solution:

(a) $\operatorname{tr}(A) = 2 + 3 + 5 = 10$.

(b)

$$A^T = \begin{pmatrix} 2 & 1 & -2 \\ -1 & 3 & 0 \\ 0 & 4 & 5 \end{pmatrix}, \qquad S = \frac{A + A^T}{2} = \begin{pmatrix} 2 & 0 & -1 \\ 0 & 3 & 2 \\ -1 & 2 & 5 \end{pmatrix}, \qquad K = \frac{A - A^T}{2} = \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & 2 \\ -1 & -2 & 0 \end{pmatrix}.$$

(c) Directly from the formulas above, $S^T = S$ and $K^T = -K$, as required of a symmetric and a skew-symmetric matrix respectively.

(d) A is neither symmetric nor skew-symmetric: from part (b), $A = S + K$ with both $S \neq 0$ and $K \neq 0$, so A does not equal its symmetric part S alone (ruling out symmetric) nor its skew-symmetric part K alone (ruling out skew-symmetric).

Problem 3

For each statement below, determine whether it is *always true*, *sometimes true*, or *never true*. Give a counterexample when appropriate.

(a) If A is symmetric, then A^2 is symmetric.

(b) If A and B are both symmetric $n \times n$ matrices, then AB is symmetric.

(c) The diagonal entries of a skew-symmetric matrix are all zero.

(d) $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$.

(e) $\operatorname{tr}(AB) = \operatorname{tr}(A)\operatorname{tr}(B)$.

Solution:

(a) **Always true.** $(A^2)^T = (AA)^T = A^T A^T = AA = A^2$ whenever $A^T = A$.

(b) **Sometimes true.** AB is symmetric iff $AB = BA$. For $A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ (both symmetric), $AB = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq (AB)^T$, so it can fail; it holds whenever A and B commute.

(c) **Always true.** If $A = -A^T$, then $A_{ii} = -A_{ii}$ for every i, forcing $A_{ii} = 0$.

(d) **Always true.** The trace is a linear operator: $\operatorname{tr}(A + B) = \sum_i (A_{ii} + B_{ii}) = \operatorname{tr}(A) + \operatorname{tr}(B)$.

(e) **Sometimes true.** Taking $A = B = I_2$ gives $\operatorname{tr}(AB) = 2 \neq 4 = \operatorname{tr}(A)\operatorname{tr}(B)$, so it is false in general, though it can coincide for special matrices (e.g. if one of them is 0).

Problem 4

Let

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}.$$

(a) Compute $y = Ax$ directly using the row-by-row definition of matrix-vector multiplication.

(b) Recompute y by writing it as a linear combination of the columns of A, weighted by the entries of x.

(c) Confirm that both methods give the same answer.

Solution:

(a) Row by row:

$$y_1 = (1)(2) + (0)(1) + (2)(-1) = 0, \qquad y_2 = (-1)(2) + (3)(1) + (1)(-1) = 0,$$

so $y = (0, 0)^T$.

(b) As a linear combination of columns $a_1 = (1, -1)^T$, $a_2 = (0, 3)^T$, $a_3 = (2, 1)^T$:

$$y = 2a_1 + 1 \cdot a_2 + (-1)a_3 = \begin{pmatrix} 2 \\ -2 \end{pmatrix} + \begin{pmatrix} 0 \\ 3 \end{pmatrix} + \begin{pmatrix} -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

(c) Both computations give $y = (0, 0)^T$, confirming the two definitions of matrix-vector multiplication agree.

Problem 5

A store chain stocks $n = 4$ products across $m = 3$ warehouses. The quantity matrix is

$$Q = \begin{pmatrix} 10 & 0 & 5 & 2 \\ 4 & 6 & 0 & 8 \\ 0 & 3 & 7 & 1 \end{pmatrix} \in \mathbb{R}^{3\times 4},$$

where Q_{ij} is the quantity of product j held at warehouse i. The unit price vector for the four products is

$$p = \begin{pmatrix} 5 \\ 10 \\ 2 \\ 7 \end{pmatrix}.$$

(a) Compute $v = Qp$, and interpret each entry of v in words.

(b) Suppose a competitor's reference stock profile is $q = (1, 1, 1, 1)^T$. Compute $Q^T w$ for some warehouse-weight vector $w = (1, 0, 0)^T$, and explain what this product represents in terms of Warehouse 1.

Solution:

(a)

$$v = Qp = \begin{pmatrix} 10(5) + 0(10) + 5(2) + 2(7) \\ 4(5) + 6(10) + 0(2) + 8(7) \\ 0(5) + 3(10) + 7(2) + 1(7) \end{pmatrix} = \begin{pmatrix} 74 \\ 136 \\ 51 \end{pmatrix}.$$

So the total value of stock is \$74 at Warehouse 1, \$136 at Warehouse 2, and \$51 at Warehouse 3.

(b) Since $w = (1, 0, 0)^T$, the product $Q^T w$ simply picks out the first column of Q^T, which is the first *row* of Q:

$$Q^T w = \begin{pmatrix} 10 \\ 0 \\ 5 \\ 2 \end{pmatrix}.$$

This is exactly the quantity of each of the four products held at Warehouse 1.

Problem 6

Let

$$u = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \qquad v = \begin{pmatrix} 2 \\ 3 \end{pmatrix}.$$

(a) Compute the outer product uv^T.

(b) Compute the Gram matrix G of the set $\{u, v\}$, i.e., the matrix whose (i, j) entry is the inner product of the i-th and j-th vectors in the list (u, v).

(c) Is G symmetric? Explain why this must always be true for a Gram matrix.

Solution:

(a)

$$uv^T = \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ -2 & -3 \end{pmatrix}.$$

(b)

$$u^T u = 2, \quad u^T v = -1, \quad v^T u = -1, \quad v^T v = 13 \quad \Longrightarrow \quad G = \begin{pmatrix} 2 & -1 \\ -1 & 13 \end{pmatrix}.$$

(c) Yes, G is symmetric: its (i, j) entry is $a_i^T a_j$ and its (j, i) entry is $a_j^T a_i$, and the inner product is symmetric ($a_i^T a_j = a_j^T a_i$), so $G_{ij} = G_{ji}$ for any Gram matrix.

Problem 7

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}, \qquad B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

(a) Compute AB and BA.

(b) Are AB and BA equal? What general property of matrix multiplication does this illustrate?

(c) Compute $(AB)^T$ and $B^T A^T$ separately, and verify that they are equal.

Solution:

(a)

$$AB = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \qquad BA = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}.$$

(b) $AB \neq BA$, illustrating that matrix multiplication is *not commutative* in general.

(c)

$$(AB)^T = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}, \qquad B^T A^T = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix},$$

so indeed $(AB)^T = B^T A^T$.

Problem 8

A two-layer neural network computes $a^{[1]} = W^{[1]}x$ and $y = W^{[2]}a^{[1]}$, where

$$W^{[1]} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}, \qquad W^{[2]} = \begin{pmatrix} 2 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

(a) Compute $a^{[1]}$ and then y.

(b) Show that y can also be written as $y = Wx$ for a single matrix W, and find W.

(c) In one or two sentences, explain why a stack of layers with no nonlinearity between them collapses into a single linear map.

Solution:

(a)

$$a^{[1]} = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix}, \qquad y = \begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ 2 \end{pmatrix} = 6.$$

(b)

$$W = W^{[2]}W^{[1]} = \begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 2 & 0 \end{pmatrix}.$$

Check: $Wx = (2)(3) + (0)(1) = 6$, matching part (a).

(c) Composing two linear maps is again a linear map: $y = W^{[2]}(W^{[1]}x) = (W^{[2]}W^{[1]})x$. With no nonlinearity in between, any number of stacked linear layers reduces to a single matrix multiplication, so the network can represent only linear functions of x regardless of depth.

Problem 9

Consider the following systems $Ax = b$. For each, write A and b, and state whether the system is under-determined, square, or over-determined; then determine whether it has no solution, a unique solution, or infinitely many solutions.

(a) $x_1 + x_2 = 3, \quad 2x_1 + 2x_2 = 6.$

(b) $x_1 + x_2 = 1, \quad x_1 - x_2 = 3, \quad 2x_1 = 5.$

(c) $x_1 + 2x_2 - x_3 = 4, \quad -x_1 + x_2 + x_3 = 0.$

Solution:

(a)
$$A = \begin{pmatrix} 1 & 1 \\ 2 & 2 \end{pmatrix}, \qquad b = \begin{pmatrix} 3 \\ 6 \end{pmatrix}.$$
This is square ($m = n = 2$), but the second equation is just twice the first, so the rows of A are linearly dependent. The system has **infinitely many solutions**: any (x_1, x_2) with $x_1 + x_2 = 3$.

(b)
$$A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \\ 2 & 0 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 3 \\ 5 \end{pmatrix}.$$
This is over-determined ($m = 3 > n = 2$). Adding the first two equations gives $2x_1 = 4$, so $x_1 = 2,\ x_2 = -1$. But then $2x_1 = 4 \neq 5$, so the system is inconsistent: **no solution**.

(c)
$$A = \begin{pmatrix} 1 & 2 & -1 \\ -1 & 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 4 \\ 0 \end{pmatrix}.$$
This is under-determined ($m = 2 < n = 3$). Letting $x_3 = t$ be free, the second equation gives $x_1 = x_2 + t$; substituting into the first gives $3x_2 = 4$, so $x_2 = 4/3$ and $x_1 = 4/3 + t$. The system has **infinitely many solutions**, parametrized by $t \in \mathbb{R}$.

Problem 10

Let
$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}.$$

(a) Is A tall, wide, or square? Could A possibly have a left inverse, a right inverse, both, or neither? Justify using the shape of A alone, before computing anything.

(b) Verify that
$$X = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}$$
is a left inverse of A by checking $XA = I$.

(c) Using X, is X also a right inverse of A? Compute AX to check.

Solution:

(a) A is tall ($m = 3 > n = 2$). A tall matrix can be left-invertible (its columns $(1,0,1)^T, (0,1,1)^T$ are not proportional, hence linearly independent), but a tall matrix can *never* be right-invertible, since right-invertibility requires the rows to be linearly independent, and three vectors in $\mathbb{R}^2$ cannot be independent.

(b)
$$XA = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$
confirming X is a left inverse of A.

(c)
$$AX = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{pmatrix} \neq I_3,$$
so X is *not* a right inverse of A, consistent with part (a).

Problem 11

Let $A \in \mathbb{R}^{m\times n}$ with $m > n$ and suppose A has linearly independent columns.

(a) Explain why A cannot be right-invertible.

(b) Write down the formula for the (left) pseudo-inverse $A^\dagger$ of A in terms of A and A^T.

(c) For

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix},$$

compute A^TA, then $A^\dagger$.

(d) Verify that $A^\dagger A = I_2$.

Solution:

(a) Right-invertibility requires the rows of A to be linearly independent. Since A has m rows lying in $\mathbb{R}^n$ with $m > n$, the rows cannot all be independent (at most n vectors in $\mathbb{R}^n$ can be independent). Hence A cannot be right-invertible.

(b) $A^\dagger = (A^TA)^{-1}A^T$.

(c) For $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$:

$$A^TA = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad (A^TA)^{-1} = \frac{1}{3}\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

$$A^\dagger = (A^TA)^{-1}A^T = \frac{1}{3}\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \end{pmatrix}.$$

(d)

$$A^\dagger A = \frac{1}{3}\begin{pmatrix} 2 & -1 & 1 \\ -1 & 2 & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = I_2.$$

Problem 12

Let $u, v, w \in \mathbb{R}^n$ be nonzero vectors and let A, B, C be matrices of compatible sizes. For each statement, determine whether it is true or false. If false, give a counterexample.

(a) If A has a left inverse, then the columns of A are linearly independent.

(b) If A is square and has a left inverse, then it also has a right inverse.

(c) A tall matrix ($m > n$) can be right-invertible.

(d) $(ABC)^T = A^TB^TC^T$.

(e) If A has orthonormal columns, then $\|Ax\| = \|x\|$ for every x.

(f) Every invertible matrix has orthonormal columns.

Solution:

(a) **True.** If $XA = I$ and $Ax = 0$, then $x = X(Ax) = X0 = 0$, so the columns of A are linearly independent.

(b) **True.** A square matrix with a left inverse has linearly independent columns, hence also linearly independent rows (equivalently, it is nonsingular), so it also has a right inverse (in fact the same matrix serves as both).

(c) **False.** A tall matrix ($m > n$) can never be right-invertible, since its $m > n$ rows cannot all be linearly independent. E.g., $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$ has no right inverse.

(d) **False.** In general $(ABC)^T = C^T B^T A^T$, not $A^T B^T C^T$. E.g., for $A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$, $C = I$, one finds $(AB)^T \neq A^T B^T$ since $AB \neq BA$ in general.

(e) **True.** If $A^T A = I$, then $\|Ax\|^2 = (Ax)^T (Ax) = x^T A^T A x = x^T x = \|x\|^2$.

(f) **False.** E.g., $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$ is invertible, but its first column has norm $2 \neq 1$, so its columns are not orthonormal.

Problem 13

A linear classifier maps a d-dimensional feature vector x to a score using a weight matrix $W \in \mathbb{R}^{k \times d}$, one row per class:

$$z = Wx, \qquad z \in \mathbb{R}^k.$$

Suppose $d = 3$, $k = 2$, and

$$W = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 2 & 1 \end{pmatrix}, \qquad x = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}.$$

(a) Compute $z = Wx$, and state which class score is higher.

(b) Viewing $z = Wx$ as a linear transformation from $\mathbb{R}^3$ to $\mathbb{R}^2$, explain in one sentence what property of the map $x \mapsto Wx$ guarantees that

$$W(x + x') = Wx + Wx', \qquad W(\alpha x) = \alpha(Wx)$$

for any vectors x, x' and scalar α.

(c) Verify the additivity property above directly using W and the two vectors $x = (2, 1, 3)^T$ and $x' = (1, 1, 0)^T$.

Solution:

(a)

$$z = Wx = \begin{pmatrix} (1)(2) + (0)(1) + (-1)(3) \\ (0)(2) + (2)(1) + (1)(3) \end{pmatrix} = \begin{pmatrix} -1 \\ 5 \end{pmatrix}.$$

Class 2's score (5) is higher than Class 1's score (-1).

(b) The map $x \mapsto Wx$ is a *linear transformation*: matrix-vector multiplication distributes over vector addition and commutes with scalar multiplication, which is precisely what guarantees additivity and homogeneity.

(c) With $x' = (1, 1, 0)^T$: $x + x' = (3, 2, 3)^T$, so

$$W(x + x') = \begin{pmatrix} (1)(3) + (0)(2) + (-1)(3) \\ (0)(3) + (2)(2) + (1)(3) \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \end{pmatrix}.$$

Also $Wx' = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, and $Wx + Wx' = \begin{pmatrix} -1 \\ 5 \end{pmatrix} + \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 7 \end{pmatrix}$, confirming $W(x + x') = Wx + Wx'$.